SBIR Phase I Technical Abstract

Project Title: Dual-Use Self-Healing Cybernetic Software Framework for Autonomous Edge Hardware and Low-Latency Robotics Orchestration

Applicant Organization: Krypticsoundz LLC

Principal Investigator: Brandon Mark Wilson

Topic Designation: Advanced Edge Computing, Robotic Autonomy, and Decentralized Systems

Technical Abstract

The reliance of modern autonomous hardware—specifically unmanned aerial vehicles (UAVs), unmanned ground vehicles (UGVs), and distributed industrial robotics—on remote cloud infrastructure introduces critical points of failure, including communication blackouts, data interception risks, and high network orchestration latencies. Krypticsoundz LLC has developed **Golden-Agent**, an enterprise-grade, self-healing cybernetic software framework designed explicitly for isolated, bare-metal edge execution. The innovation completely decouples autonomous software orchestration from external network dependencies by utilizing a unique hybrid architecture. High-speed, native C++17 compile-time extensions are bound directly within an agile Python telemetry loop, achieving a verified, sub-millisecond (sub-1.0 ms) local processing loop density.

Furthermore, the system implements continuous hardware-aware resource regulation by reading directly from physical Linux `/sys/class/` paths, allowing real-time throttling of execution cycles based on live thermal variations and battery depletion metrics. Data integrity is enforced through a zero-trust cryptographic persistence layer backed by SHA-256 verification matrices to ensure secure memory state migration across operational crashes.

Under the Phase I effort, Krypticsoundz LLC will focus on environment hardening, optimizing the C++ bridge throughput, and integrating automated code-patching modules to isolate system faults dynamically. The final milestone will conclude with a field-ready hardware prototype demonstration validating performance density under simulated tactical shocks.

Commercial Capability & Dual-Use Impact

The framework addresses high-priority gaps across both defense and civilian technical ecosystems. In the defense sector, the system provides standalone, hyper-privileged process isolation for autonomous drone payloads operating in contested, zero-connectivity electronic warfare environments. Commercially, the software can be white-labeled and integrated into industrial assembly automation, deep-sea exploration vessels, remote energy infrastructure monitoring, and high-value medical diagnostic hardware that require zero-fail local processing and robust physical data security.